Implicit Differentiation

Exam code: 9MA0

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Implicit differentiation

What is implicit differentiation?

- An equation connecting ***x*** and ***y*** is not always easy to write **explicitly** in the form ***y*= f(*x*)** or ***x* = f(*y*)**
- However you can still differentiate such an equation **implicitly** using the **chain rule:**

$$1.\quad \frac{d}{dx} f(y) = f'(y)\frac{dy}{dx}$$

- $\frac{d}{dx}$ MEANS "THE DERIVATIVE WITH RESPECT TO x (OF WHAT FOLLOWS)". SO $\frac{d}{dx}f(y)$ HERE IS DERIVATIVE OF f(y) WITH RESPECT TO x.
- $f'(y)$ IS THE DERIVATIVE OF f(y) WITH RESPECT TO y.
- IT IS IMPORTANT TO KEEP THIS STRAIGHT! (SEE THE EXAMPLE BELOW).

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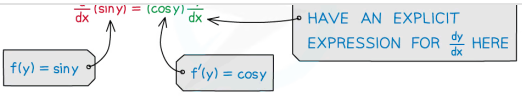
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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- Combining this with the **product rule** gives us:

$$2.\quad \frac{d}{dx}(f(x)g(y)) = f'(x)g(y) + f(x)g'(y)\frac{dy}{dx}$$

- $f'(x)$ IS THE DERIVATIVE OF $f(x)$ WITH RESPECT TO x.
- $g'(y)$ IS THE DERIVATIVE OF $g(y)$ WITH RESPECT TO y.

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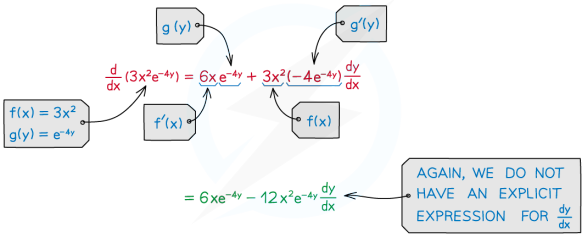
e.g. FIND THE DERIVATIVE OF $3x^2e^{-4y}$ WITH RESPECT TO x.



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- These two special cases are especially useful:

$$3.\quad \frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$$

SPECIAL CASE OF 1. WITH $f(y) = y^n$

$$4.\quad \frac{d}{dx}(xy) = x\frac{dy}{dx} + y$$

SPECIAL CASE OF 2. WITH $f(x) = x$ AND $g(y) = y$

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- When ***x*** and ***y*** are connected in an **equation** you can differentiate both sides with respect to ***x*** and rearrange to find a formula (usually in terms of ***x* and *y***) for **dy/dx**
  - Note that **dy/dx** is a single algebraic object
  - When rearranging do not treat **dy/dx** as a fraction
  - Especially do not try to separate **dy** and **dx** and treat them as algebraic objects on their own!

USE IMPLICIT DIFFERENTIATION TO FIND AN EXPRESSION FOR $\frac{dy}{dx}$.

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}(3x^2e^{-4y})$$

DIFFERENTIATE BOTH SIDES WITH RESPECT TO x

$$\cos y\frac{dy}{dx} = 6xe^{-4y} - 12x^2e^{-4y}\frac{dy}{dx}$$

WE ALREADY WORKED OUT THESE DERIVATIVES ABOVE!

$$(12x^2e^{-4y} + \cos y)\frac{dy}{dx} = 6xe^{-4y}$$

REARRANGE TO MAKE $\frac{dy}{dx}$ THE SUBJECT

$$\frac{dy}{dx} = \frac{6xe^{-4y}}{12x^2e^{-4y} + \cos y}$$

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Examiner Tips and Tricks

- When using implicit differentiation you will not always be able to write **dy/dx** simply as a function of ***x***.
- However, this does not stop you from answering questions involving the derivative.

Worked Example

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